Hermes: Towards Bridging Relational and Cryptographic Abstractions in Homomorphically Encrypted Databases

Anonymous Author(s)

Abstract

Fully Homomorphic Encryption (FHE) promises the ability to compute over encrypted data without revealing sensitive contents. Yet, enabling this feature inside real-world relational database systems remains elusive due to prohibitive performance overhead and the structural mismatch between mutable database tuples and cryptographically immutable FHE ciphertexts.

This paper presents Hermes, a prototype system that enables homomorphically encrypted queries directly on a relational database system, e.g., MySQL. To amortize FHE costs, Hermes introduces a new SIMD-aware data model beyond simply packing vectorized tuples within a ciphertext: precomputed aggregate statistics becomes a first-class citizen and is dynamically maintained within the ciphertext to support efficient aggregations without expensive cryptographic rotations, i.e., Galois automorphisms. Furthermore, to support mutable ciphertexts, we develop data-oblivious homomorphic algorithms built upon polynomial slot masking and shifting, which we show provably secure under standard security models.

Hermes has been implemented as a series of C++ loadable functions in MySQL, marking the first practical integration of FHE into an production database system. Extensive evaluations across diverse datasets demonstrate an over 3400x increase in encryption throughput, an over 4000x speedup for tuple insertions, and a 300x acceleration for deletions when compared to conventional FHE implementations.

CCS Concepts

• **Security and privacy** → **Cryptography**; • **Information systems** → **Database management systems**.

Keywords

Database security, Fully homomorphic encryption, User-defined functions

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1 Introduction

1.1 Background and Motivation

For over four decades, database systems have formed the backbone of modern data infrastructure, enabling structured storage, efficient retrieval, and scalable analytics. As the industry has transitioned to cloud-first architectures, the confidentiality of outsourced data has become a central concern. Increasingly, users demand end-to-end cryptographic protection throughout the entire computation pipeline, rather than merely during transmission or at rest. Among cryptographic primitives, Fully Homomorphic Encryption (FHE) offers a mechanism to evaluate algebraic operations directly on encrypted data, removing the requirement to trust the compute server and, argubally more importantly, precluding internal attackers from within the cloud vendor. Following Gentry's theoretical (yet impractical due to the prohibitive computational costs) framework [19], subsequent schemes such as BFV and CKKS [11, 17] have improved computational efficiency and implementation maturity [3]. FHE is now applied in practical domains, including secure machine learning and privacy-preserving analytics [7, 23].

However, (relational) database systems have yet to fully integrate FHE. Although extensive research exists on encrypted databases [35, 36, 43], operating a relational engine directly on semantically secure ciphertexts remains an open challenge. One of the most notable issues stems from a structural mismatch between database semantics and cryptographic notion. Traditional database management systems are designed for mutable records, evolving schemas, and runtime query execution plans. In contrast, FHE schemes operate over static circuits and immutable (yet randomized, informally speaking) ciphertexts. Resolving this structural divide requires more than wrapping FHE function calls within SQL. It necessitates a sort of bridge that maps the operational dynamics of relational execution to the algebraic invariants of cryptographic computation. Without bridging these abstractions, applying cryptographic schemes to secure query processing remains impractical.

1.2 Proposed Work

We present *Hermes*, a prototype system that enables secure and efficient vectorized relational query execution over homomorphically encrypted data. Built atop the multi-slot capabilities of modern FHE schemes, Hermes is implemented as a suite of native MySQL loadable functions interfacing with OpenFHE [3]. Hermes bridges the gap between relational and cryptographic abstractions by introducing four key innovations:

(1) Packed Relational Data Model. Hermes adopts a packed ciphertext representation that maps multiple database tuples into parallel homomorphic slots. To break the performance bottleneck of homomorphic analytics, we design a physical layout that reserves a specific slot within each ciphertext. This dedicated slot stores a precomputed auxiliary value representing the local sum of the data

slots. By embedding this analytic state directly into the physical storage layout, Hermes establishes a foundation for efficient query execution that completely bypasses the need for costly Galois automorphisms. We elaborate on the design of this static data model in Section 3.1.

(2) Data-Oblivious Mutability Primitives. To allow a single ciphertext to reflect the updates of mutable records, Hermes supports dynamic record-level modifications directly on packed ciphertexts. Insertions and deletions are executed via homomorphic shift-and-mask operations. Inserting a new record shifts a tail segment rightward, while deleting a record shifts the segment leftward to clear the target slot. These structural updates are performed using homomorphic masking and slot-wise addition without requiring decryption or plaintext recovery. Such operations preserve semantic security and enable efficient in-place mutability while keeping the database access patterns completely opaque to the server. We present the design of these primitives in Section 3.2.

(3) Homomorphic Aggregation and Maintenance. Building upon the packed layout and mutability primitives, Hermes introduces a closed-loop mechanism for analytical queries. Encrypted aggregation queries such as `SUM` can be answered using constant-time ciphertext additions of the reserved auxiliary slots. When data mutations occur, Hermes synchronously updates this auxiliary slot using lightweight homomorphic scalar operations. This dual approach amortizes the computational overhead of fully homomorphic encryption and drastically improves overall query throughput. We discuss this execution pipeline in Section 3.3.

(4) Native Database Integration and Rigorous Analysis. Hermes facilitates a full-stack encrypted query execution pipeline natively within MySQL. The system is realized as a collection of C++ modules registered via the MySQL loadable function interface, which avoids client-side orchestration and minimizes ciphertext transformations. Furthermore, we provide a comprehensive theoretical framework proving the IND-CPA security of our mutability primitives and formally establishing the sub-linear asymptotic complexity of our operations. The theoretical analysis and system implementation are detailed in Section 3.4 and Section 4.

Our extensive evaluations in Section 5 are conducted across diverse real-world datasets, encompassing genomic annotations, cryptocurrency trade volumes, and healthcare statistics. We comprehensively benchmark the system against four distinct execution baselines: unencrypted plaintext SQL, conventional singular FHE, a standard packed FHE configuration relying on logarithmic Galois automorphisms [3], and the Intel Paillier Cryptosystem Library [26]. Extensive evaluations demonstrate significant performance improvements, including an over 3400x increase in encryption throughput, a 4000x speedup for tuple insertions, and a 300x acceleration for deletions when compared to conventional scalar FHE deployments.

2 Related Work and Preliminaries

2.1 Encrypted Databases

One major class of encrypted databases relies on client-side processing, where query execution is deferred until the encrypted data is downloaded to a trusted local environment [1]. While this architecture strictly minimizes trust assumptions by keeping all analytical logic away from the cloud provider, it inherently sacrifices the computational benefits of server-side scalability. Transferring large volumes of encrypted records across the network incurs severe bandwidth penalties and high latency, effectively reducing the cloud database management system to a passive storage repository rather than an active execution engine.

An alternative approach attempts to split query execution between the client and the server, as seen in earlier information-hiding systems [22]. In this model, server-side filters retrieve encrypted tuples guided by secure indices or cryptographic tokens generated by the client. Building upon this paradigm, systems like CryptDB [36], Arx [35], and more recently ones [10, 16, 18, 32, 37, 40–42, 45–47, 52, 54, 57, 60, 61] explore different structural trade-offs between query functionality and data leakage. They employ layered encryption onions, partitioned data encodings, or specialized cryptographic circuits to evaluate relational predicates. However, these designs frequently leak access patterns or require multiple rounds of network interaction, as the server must often return intermediate candidate sets for the client to decrypt, refine, and re-upload.

To mitigate the computability over encrypted data, homomorphic encryption has been explored to integrate into encrypted database systems to compute over ciphertexts without continuous client intervention. For instance, Symmetria [38] utilizes multiplicative homomorphic encryption combined with additive extensions, whereas Rache [43] leverages layout-aware ciphertext placement alongside SIMD-optimized key switching techniques; FHE has also been applied to specific queries, such as [27, 31, 34, 51, 55]. Additional efforts from the applied cryptography community, such as HE3DB [5] and ArcEDB [59], introduce new arithmetic and logic operators to accelerate SQL evaluations over ciphertexts. However, these solutions often manifest as standalone programmatic scripts rather than being fully integrated into a standard relational database engine. Furthermore, to mitigate the extreme performance penalties and noise growth of FHE, recent query engines like NSHEDB [28] rely on word-level leveled homomorphic encryption. Despite these cryptographic advances, existing homomorphic databases typically operate on scalar records, encrypting individual database attributes into isolated ciphertexts. This scalar approach fails to exploit the parallel processing capabilities of modern FHE schemes, leading to storage inflation and computational bottlenecks during query execution.

Hermes differs from prior systems by integrating homomorphic encryption directly into the ingestion path. Unlike approaches that rely on client-side ciphertext generation or immutable payload storage, Hermes exposes SQL-level UDFs for packing, encrypting, and mutating ciphertexts at runtime. Our system introduces a slot-aware data model that supports in-place ciphertext updates and maintains per-group local aggregates as part of the packing process.

2.2 Fully Homomorphic Encryption

Homomorphic encryption (HE) enables computation on encrypted data without decryption. The seminal work by Gentry [19] introduced the first fully homomorphic encryption (FHE) scheme, laying the foundation for later constructions such as BFV [8, 17],

GSW [20], BGV [9], and CKKS [11]. TFHE [12] supports Boolean circuits and has seen recent improvements in bootstrapping performance [21, 44]. These schemes have been implemented in libraries including SEAL [39], HElib [24], OpenFHE [3], and LattiGo [30].

Hermes is built atop the BFV scheme [17], a lattice-based fully homomorphic encryption (FHE) construction supporting exact arithmetic over integers modulo a plaintext modulus t. Let $R = \mathbb{Z}[X]/(X^N+1)$ be a cyclotomic ring of degree $N = 2^k$ for some $k \in \mathbb{N}$. A plaintext message is represented as an element of $R_t := R/tR$, while ciphertexts reside in R_q^2, where $q \gg t$ is a large ciphertext modulus.

A key feature of BFV is its support for *plaintext packing* via the Chinese Remainder Theorem (CRT). When $t \equiv 1 \mod 2N$, the ring R_t splits as a direct product of $n = N/2$ slots:

$$R_t \cong \mathbb{Z}_t^n,$$

enabling SIMD-style evaluation of vectorized inputs. Each plaintext $\mathbf{m} = (m_0, \ldots, m_{n-1}) \in \mathbb{Z}_t^n$ is encoded into a single polynomial $m(X) \in R_t$, encrypted as $\mathsf{Enc}(m) = \mathbf{c} \in R_q^2$, and evaluated homomorphically across all slots.

Homomorphic operations preserve slot-wise semantics:

$$\mathsf{EvalAdd}(\mathsf{Enc}(\mathbf{m}), \mathsf{Enc}(\mathbf{m}')) = \mathsf{Enc}(\mathbf{m} + \mathbf{m}'),$$
$$\mathsf{EvalMult}(\mathsf{Enc}(\mathbf{m}), \mathsf{Enc}(\mathbf{m}')) = \mathsf{Enc}(\mathbf{m} \cdot \mathbf{m}'),$$

where the arithmetic is component-wise modulo t. In addition, the BFV scheme supports automorphism-based rotation operations, also known as *Galois rotations*, which cyclically shift slot positions:

$$\mathsf{EvalRotate}(\mathsf{Enc}(\mathbf{m}), r) = \mathsf{Enc}((m_{(i+r) \bmod n})_{i=0}^{n-1}).$$

Hermes leverages this packing structure to encode multiple database tuples into a single ciphertext. This design enables parallel evaluation, slot-wise updates, and sublinear aggregation. Importantly, all packing and update logic remains entirely within the encrypted domain, without leaking intermediate values or structural metadata.

In addition, the BFV scheme supports automorphism-based slot rotation using Galois group actions. For any $r \in \mathbb{Z}$, a cyclic slot rotation is realized via the Galois automorphism $\sigma_r : X \mapsto X^r \mod (X^N+1)$, which acts on ciphertexts by permuting their internal slot layout. To evaluate this transformation homomorphically, the client must precompute and upload the corresponding Galois keys:

$$\mathsf{EvalAtIndex}(\mathsf{Enc}(\mathbf{m}), r) = \mathsf{Enc}((m_{(i+r) \bmod n})_{i=0}^{n-1}),$$

where the rotation index r corresponds to a specific automorphism in the Galois group $\mathrm{Gal}(R_q/\mathbb{Q})$. These keys enable Hermes to perform selective slot access, masking, and reorganization within encrypted vectors while preserving semantic security.

2.3 Alternative Secure Processing Paradigms

Alongside purely cryptographic storage, trusted hardware enclaves provide an alternative route for secure data processing. Systems utilizing Intel SGX such as ObliDB [15] and CryptSQLite [49] protect data in memory while offering oblivious query processing to prevent access pattern leaks. Enterprise solutions have also adopted this paradigm. For instance, recent iterations of Azure SQL Database [2] leverage enclaves to enable rich pattern matching over encrypted columns. Furthermore, hardware isolation techniques have been proposed to secure peripheral interfaces and web environments against side channel attacks [14]. While enclaves deliver high performance, they introduce strict hardware dependencies and remain vulnerable to sophisticated microarchitectural exploits, which pure homomorphic encryption avoids.

Another prominent paradigm involves Secure Multi-party Computation and federated privacy. Frameworks like Conclave [48] and Shrinkwrap [4] accelerate analytics over big data by partitioning queries between cleartext local processing and secure protocols. Similarly, advanced join aggregate queries can be evaluated across private datasets without exposing individual records [50]. Managing cross cryptographic leakages in such distributed databases is a complex endeavor [58]. Additionally, massive parallel query optimization strategies [53] and Private Information Retrieval [56] remain essential for scalable data extraction. However, multi-party protocols inherently require heavy network synchronization among non-colluding servers, which is a fairly strong assumption that may or may not hold for many practical scenarios. Hermes bypasses these communication bottlenecks by shifting the computational burden entirely to the server side evaluation of SIMD packed ciphertexts.

2.4 Terminology and Notation

We summarize key terms and notation used throughout this paper. Let n denote the number of slots supported by the plaintext modulus and ring parameters of the BFV encryption scheme. All vectors are implicitly assumed to be over $\mathbb{Z}_t^n$, where t is the plaintext modulus.

Slot. Each ciphertext encodes a vector of n plaintext elements, known as *slots*. Slots are independently addressable under SIMD operations supported by BFV.

Group ID. In Hermes, input tuples are partitioned by a logical *group ID*, which assigns each record to a packed vector instance. All records within the same group are encoded into the same ciphertext. The mapping from record to group is deterministic and index-driven.

Auxiliary Sum Slot. Hermes reserves the final slot (slot $n-1$) in each ciphertext to store the *local sum*, i.e., the plaintext sum $\sum_{j=0}^{n-2} m_j$ of all payload slots. This value is computed at packing time and used to accelerate encrypted aggregation without Galois rotation.

Packed Representation. A ciphertext that stores $n-1$ application values and one auxiliary statistic is referred to as a *packed representation*. This format enables efficient updates and aggregation via SIMD-aware primitives.

Logical Length. Some plaintext vectors do not occupy all n slots. The number of meaningful entries is called the *logical length*, and is used to control serialization, summation, and slot masking operations.

3 Hermes: Native FHE for Relational Databases

3.1 The Hermes Packed Data Model

To bridge the structural gap between relational database tables and cryptographic objects, we propose a slot-aware packing layout. In

this model, each ciphertext acts as a secure container holding a fixed-size group of database records. This architecture amortizes the encryption cost and establishes the foundation for efficient SIMD-style query processing over the encrypted domain.

3.1.1 Slot-Aware Packing Strategy. In Hermes, we introduce a structured packing layout that leverages the SIMD-style slot architecture of the BFV homomorphic encryption scheme. Rather than encrypting each tuple into an isolated ciphertext, which incurs prohibitive memory and performance overhead, we adopt a batched layout where each ciphertext encodes a vector of plaintext values, referred to as slots.

Let N denote the ring dimension of the BFV scheme. When batching is enabled via the Chinese Remainder Theorem on cyclotomic rings, the scheme yields $n = N/2$ usable slots. For example, with $N = 8192$, Hermes can store $n = 4096$ individual scalar values per ciphertext. However, instead of utilizing all n slots for raw data, we strictly reserve the final slot to store an auxiliary value, such as the local sum of the vector entries. This yields an effective capacity of $n - 1$ logical records per ciphertext.

Each slot in a packed ciphertext has a fixed positional meaning: slot i corresponds to the i-th logical record in the group. This deterministic alignment enables precise slot-wise reasoning. When a ciphertext is decrypted, the unpacked vector reveals values at well-defined offsets, allowing the database client to reconstruct the plaintext table fragment without additional metadata. This position-indexed design is essential for supporting secure in-place modifications without leaking tuple identity or requiring explicit key-column mapping.

3.1.2 Auxiliary Slot Embedding at Packing Time. Hermes adopts a hybrid encoding strategy in which each packed ciphertext stores not only the array of plaintext relational values but also a precomputed auxiliary statistic. This local sum is embedded into the last slot of the ciphertext at the exact time of ingestion and packing.

Given a batch of $n - 1$ plaintext values $(v_0, v_1, \ldots, v_{n-2})$ to be packed into a ciphertext, the system computes their sum $s = \sum_{i=0}^{n-2} v_i$ in plaintext and constructs an extended vector of the form:

$$m = (v_0, v_1, \ldots, v_{n-2}, s) \in \mathbb{Z}_t^n,$$

where t is the plaintext modulus. The extended vector m is then passed to the cryptographic API to generate a packed plaintext object, which is subsequently encrypted into a ciphertext.

This layout effectively reserves the final slot at index $n - 1$ as a local accumulator. Because BFV ciphertexts support SIMD-style evaluations, all data and the auxiliary sum coexist securely within a single ciphertext boundary. Since the sum is computed in the clear before encryption, this initial embedding incurs no homomorphic overhead. By formalizing this static data model, Hermes transforms complex multi-tuple structures into manageable relational vectors, setting the stage for the dynamic mutability primitives discussed in the next section.

3.2 Data-Oblivious Mutability Primitives

To allow a single packed ciphertext to dynamically reflect the updates of mutable database records, Hermes supports record-level modifications directly within the encrypted domain. Unlike conventional systems that require downloading, decrypting, and re-encrypting the entire ciphertext for a single update, our design ensures that mutability is achieved in-place. We design a homomorphic slot manipulation interface that incorporates plaintext masking, Galois shifting, and slot-wise additions.

3.2.1 Homomorphic Insertion. Hermes supports the in-place insertion of plaintext values into packed ciphertexts. The objective is to insert a new value v into logical slot i of an existing ciphertext c, where slots i through $n - 2$ may already contain payload data. Since ciphertexts are encoded as fixed-size relational vectors, this operation requires shifting all subsequent entries rightward by one position. This is analogous to an array insertion, yet it is performed entirely under encryption.

The procedure utilizes homomorphic Galois rotations and plaintext masks. We first isolate the tail of the vector using a preservation mask and shift it rightward. Concurrently, we isolate the head of the vector to keep the preceding records intact. The new value is then encrypted and injected into the target slot. Finally, the auxiliary local sum at the end of the ciphertext is synchronously updated via a scalar plaintext addition.

We formalize this procedure in Algorithm 1.

Algorithm 1: Data-Oblivious Homomorphic Insertion

Require: Ciphertext c, Target index i, Plaintext v, Capacity n
Ensure: Updated Ciphertext c_{new}

$m_{head} \leftarrow$ vector with 1s in $[0, i-1]$ and 0s elsewhere
$m_{tail} \leftarrow$ vector with 1s in $[i, n-3]$ and 0s elsewhere
$c_{head} \leftarrow \mathsf{EvalMult}(c, m_{head})$
$c_{tail} \leftarrow \mathsf{EvalMult}(c, m_{tail})$
$c_{shifted} \leftarrow \mathsf{EvalRotate}(c_{tail}, 1)$
$m_v \leftarrow$ vector with v at index i and 0s elsewhere
$c_v \leftarrow \mathsf{Encrypt}(m_v)$
$c_{payload} \leftarrow \mathsf{EvalAdd}(c_{head}, \mathsf{EvalAdd}(c_{shifted}, c_v))$
$m_{sum} \leftarrow$ vector with v at index $n-1$ and 0s elsewhere
$c_{new} \leftarrow \mathsf{EvalAdd}(c_{payload}, m_{sum})$
return c_{new}

This operation is executed entirely within the database server process using the OpenFHE API. It leaks no information regarding which slot was modified or what value was inserted, maintaining a completely data-oblivious execution profile. When a ciphertext reaches its maximum slot capacity, Hermes rejects further insertions, maintaining a strict one-to-one mapping between the database group and the ciphertext.

3.2.2 Homomorphic Deletion. Hermes enables encrypted deletion by simulating a logical left-shift over the underlying slots. The goal is to remove the value at slot i and maintain a compacted layout where all subsequent values are shifted leftward to fill the void. The final payload slot is then cleared.

The deletion procedure follows a symmetric logic to insertion. Hermes isolates the subvector corresponding to slots $i + 1$ through $n - 2$ and rotates it leftward by one position. A zeroing mask is subsequently applied to clear the final payload slot, which would otherwise contain a stale duplicate value due to the shift. The

original head of the vector is preserved and merged with the shifted tail. The local sum is then decremented by the deleted value via plaintext subtraction.

Algorithm 2 details the formal steps for this operation.

Algorithm 2: Data-Oblivious Homomorphic Deletion

Require: Ciphertext c, Target index i, Plaintext v, Capacity n
Ensure: Updated Ciphertext c_{new}

$m_{head} \leftarrow$ vector with 1s in $[0, i-1]$ and 0s elsewhere
$m_{tail} \leftarrow$ vector with 1s in $[i+1, n-2]$ and 0s elsewhere
$c_{head} \leftarrow \mathsf{EvalMult}(c, m_{head})$
$c_{tail} \leftarrow \mathsf{EvalMult}(c, m_{tail})$
$c_{shifted} \leftarrow \mathsf{EvalRotate}(c_{tail}, -1)$
$z \leftarrow$ vector with 0 at index $n-2$ and 1s elsewhere
$c_{clean} \leftarrow \mathsf{EvalMult}(c_{shifted}, z)$
$c_{payload} \leftarrow \mathsf{EvalAdd}(c_{head}, c_{clean})$
$m_{sum} \leftarrow$ vector with $-v$ at index $n-1$ and 0s elsewhere
$c_{new} \leftarrow \mathsf{EvalAdd}(c_{payload}, m_{sum})$
return c_{new}

All steps are performed homomorphically, ensuring slot occupancy consistency and maintaining statistical privacy. If all payload slots are deleted, the ciphertext is treated as semantically inert. It contributes nothing to aggregate queries, and the auxiliary sum naturally evaluates to zero, guaranteeing that global aggregation remains perfectly accurate even in the presence of entirely empty groups.

3.3 Homomorphic Aggregate and Maintenance

Building upon the static packed layout and the dynamic mutability primitives, Hermes introduces a closed loop mechanism for evaluating analytical queries over encrypted data. This section describes how the system executes global aggregations and how it incrementally maintains the necessary auxiliary statistics to guarantee query correctness.

3.3.1 Encrypted Global Aggregation. Traditional approaches to encrypted aggregation require expensive slot wise traversal and sequential Galois rotations to accumulate values within a packed ciphertext. Hermes bypasses this computational bottleneck by leveraging the precomputed local sum embedded in the final slot of each ciphertext.

For global analytical queries such as `SELECT SUM(attr)`, the database engine does not need to unpack or rotate the individual data slots. Instead, the query planner translates the aggregation into a single ciphertext level addition. Given a collection of ciphertexts corresponding to a queried table or group, Hermes executes a homomorphic addition across all ciphertexts. Because the BFV scheme operates in a SIMD fashion, this single addition simultaneously aggregates the payload slots and the auxiliary sum slots. The database client then decrypts the aggregated result and strictly reads the value at the final slot $n-1$. This transforms an operation that typically requires logarithmic rotation depth into a constant time ciphertext addition, drastically reducing the latency of encrypted analytics.

3.3.2 Lightweight Incremental Maintenance. To ensure the correctness of aggregation queries over mutable data, the embedded local sum must remain perfectly synchronized with the active payload slots. Whenever an insertion or deletion occurs, Hermes performs a lightweight incremental update on the auxiliary slot rather than recomputing the sum from scratch.

As demonstrated in the mutability primitives, when a new plaintext value v is inserted into a packed ciphertext, Hermes isolates the update to the auxiliary slot by constructing a sparse plaintext vector. This vector contains the value v exactly at index $n-1$ and zeros elsewhere. The system then performs a homomorphic scalar addition to increase the local sum to reflect the newly ingested record. Conversely, when a record is deleted, Hermes constructs a similar sparse vector containing the negated value $-v$ at the final slot to decrement the local sum. Because the inserted or deleted values are supplied in cleartext as part of the SQL update expression, these incremental adjustments require no re-encryption or multi level circuit evaluations. We formalize both the global aggregation procedure and the incremental maintenance logic in Algorithm 3.

Algorithm 3: Homomorphic Aggregate and Maintenance

Require: Ciphertexts $c_1, \ldots, c_k$, Target c, Update v, Capacity n
Ensure: Aggregated result c_{agg}, Maintained ciphertext c_{new}

// Part 1: Accelerated Global Summation
$c_{agg} \leftarrow c_1$
for each subsequent ciphertext c_j $(j = 2 \ldots k)$ **do**
  $c_{agg} \leftarrow \mathsf{EvalAdd}(c_{agg}, c_j)$ ▹ Aggregate of auxiliary slots
end for
// Part 2: Incremental Maintenance of Local Sum
Create sparse plaintext vector $m_{sum} \in \mathbb{Z}_t^n$
Set $m_{sum}[n-1] \leftarrow v$ ▹ Align with the auxiliary slot index
$c_{new} \leftarrow \mathsf{EvalAdd}(c, m_{sum})$ ▹ Scalar update to accumulator
return c_{agg} and c_{new}

3.4 Security and Complexity Analysis

3.4.1 Security Model and Assumptions. We analyze the security of Hermes under the standard indistinguishability under chosen-plaintext attack (IND-CPA) model, instantiated using BFV [17]. We assume a passive, honest-but-curious adversary with full access to encrypted database contents, including ciphertexts before and after updates, as well as query results. The adversary may observe ciphertext structure, slot layout, and numerical patterns in decrypted outputs but is assumed to follow protocol without actively deviating from the query logic. However, the server does not record low-level timing, memory access, or power traces: Preventing side-channel attacks is beyond the scope of this work.

In order to ensure that Hermes does not introduce any new leakage channels beyond what is inherent in the underlying FHE scheme, we require the following invariants to hold:

(1) *Indistinguishability of Ciphertexts:* For any two payload vectors v and v' of the same length, their packed ciphertexts $c = \mathsf{Encrypt}(v)$ and $c' = \mathsf{Encrypt}(v')$ are computationally indistinguishable under the IND-CPA security of BFV.

(2) *Update Pattern Privacy:* Insertion and deletion operations do not reveal the affected slot position or the identity of the modified value to the server. That is, for any two slots i and j, the adversary cannot distinguish an insertion at slot i from one at j by examining ciphertext differences alone.
(3) *Aggregation Consistency:* Global aggregation over auxiliary slots (e.g., encrypted 'SUM') reveals no more information than the final aggregate result; the per-ciphertext local sum values are protected under the encryption scheme.
(4) *No Adaptive Leakage:* Hermes does not reveal any intermediate plaintexts or randomness that could be used to link ciphertexts over time. All operations are performed homomorphically and reuse no client secrets post-encryption.

Importantly, the Hermes data model and update interface preserve these properties without relying on secure multi-party computation (MPC), secure enclaves, or trusted client-side computation. All operations, such as packing, insert/delete, and aggregation, are fully executed on encrypted data within the untrusted database server.

This model captures realistic cloud deployment scenarios where users outsource encrypted databases to an untrusted cloud provider but wish to retain strong confidentiality guarantees under standard cryptographic assumptions (e.g., Ring-LWE hardness). We note that this threat model is stronger than many property-preserving encryption (PPE) systems, which often leak order, frequency, or access pattern by design.

3.4.2 Preservation of IND-CPA Security. The semantic security of Hermes relies on the assumption that the underlying BFV scheme remains IND-CPA secure under all ciphertext manipulations performed in the system. This includes ciphertext packing, masked updates, homomorphic insertion and deletion, and the embedding of auxiliary local statistics such as the group sum. We now argue that Hermes does not introduce any structural leakage or functional dependencies that could be exploited to distinguish ciphertexts. The formal proof is based on the following two observations.

First, although Hermes embeds deterministic metadata (e.g., local sums) into specific slots, the entire ciphertext is encrypted using randomized BFV encryption over the full plaintext vector. IND-CPA security guarantees that the ciphertext remains indistinguishable even if some slots are derived from others, so long as the message vector is encrypted as a whole with fresh randomness. In particular, the inclusion of auxiliary statistics does not compromise the semantic security of any individual slot.

Second, Hermes performs all slot-level updates, such as insertion and deletion, purely through homomorphic operations. These updates use masking, shifting, and slot-wise additions, but never invoke decryption, re-encryption, or key switching. No new randomness is injected, and no plaintext values are revealed or reconstructed during runtime. As a result, each modified ciphertext maintains the same IND-CPA security level as its original version, with no additional leakage beyond the allowed ciphertext interface.

We start by recalling the notion of IND-CPA security.

Definition 3.1 (IND-CPA Security). Let $\mathsf{Enc}_{\mathsf{pk}}(m; r)$ denote the encryption of a plaintext m under public key pk and randomness r. A scheme is IND-CPA secure if no probabilistic polynomial-time (PPT) adversary $\mathcal{A}$ can distinguish between encryptions of m_0 and m_1 for any chosen pair of equal-length plaintexts, even given arbitrary access to the encryption oracle. Formally, for any PPT adversary $\mathcal{A}$, the advantage

$$\mathrm{Adv}_{\mathcal{A}}^{\text{IND-CPA}} = |\Pr[\mathcal{A}(\mathsf{Enc}(m_0)) = 1] - \Pr[\mathcal{A}(\mathsf{Enc}(m_1)) = 1]|$$

is negligible in the security parameter λ.

Lemma 3.2. *Let C be the set of ciphertexts generated and manipulated by Hermes. Then under the assumption that the base FHE scheme (BFV) is IND-CPA secure, the ciphertext $c \in C$ remains IND-CPA secure throughout all operations in Hermes.*

Proof Sketch. The proof relies on a reduction to the underlying IND-CPA security of the BFV scheme. Since Hermes constructs its packed data model using standard plaintext encoding and standard homomorphic evaluations, any adversary capable of distinguishing between two Hermes ciphertexts could be leveraged to break the semantic security of the base BFV encryption.

Furthermore, the deterministic inclusion of auxiliary local sums and the application of public plaintext masks during updates do not introduce new leakage channels. All slot modifications are performed strictly within the encrypted domain without relying on secret-dependent control flows. Due to space constraints, the full formal reduction is deferred to our extended technical report.

The full proof can be found in the Appendix, where we construct a simulator that interacts with the IND-CPA challenger for BFV and demonstrates that any advantage in distinguishing Hermes ciphertexts translates to an advantage against the base scheme, thus confirming that Hermes maintains IND-CPA security under the given assumptions. □

An important subtlety in Hermes is the deterministic embedding of auxiliary data into a fixed slot. While this creates a structural invariant, it does not violate IND-CPA since the data being embedded is functionally independent on other encrypted slots. From a cryptographic perspective, this corresponds to publishing a function $f(v_0, \ldots, v_{n-2}) = s$ along with the encryption of the vector, which is a standard practice in FHE-based secure computation. Furthermore, all masking and slot operations are purely homomorphic and do not involve conditional branching or secret-dependent control flow. No part of the computation leaks information via side channels, timing, or slot access patterns.

3.4.3 Update Privacy via Slot-Level Obfuscation. To preserve data obliviousness within packed ciphertexts, Hermes ensures that individual insertions and deletions are indistinguishable with respect to their slot index. Since encrypted updates are performed via uniform homomorphic operations, namely Galois rotations, plaintext masking, and slot-wise additions, the adversary cannot determine which slot was modified given a before-and-after pair of ciphertexts. We formalize this guarantee using a slot-level indistinguishability game.

We start by formally defining the game.

Definition 3.3 (Slot Update Indistinguishability Game). Let c be an encrypted packed ciphertext over n slots under BFV. Consider two update positions $i_0, i_1 \in \{0, \ldots, n-2\}$. The goal of the adversary $\mathcal{A}$ is to distinguish whether an insertion was applied at position i_0 or i_1. The game proceeds as follows:

(1) Challenger generates keypair $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^\lambda)$.
(2) Adversary chooses a plaintext vector $\vec{v} = (v_0, \ldots, v_{n-2})$ and computes $c \leftarrow \mathsf{Enc}_{\mathsf{pk}}(\vec{v})$.
(3) Challenger chooses a random bit $b \in \{0, 1\}$.
(4) Challenger inserts a known value δ at slot i_b using Hermes' homomorphic insert logic, resulting in $c' = \mathsf{Insert}(c, i_b, \delta)$.
(5) Adversary is given c' and outputs a guess $b' \in \{0, 1\}$.

We define the adversary's advantage as:

$$\mathrm{Adv}_{\mathcal{A}}^{\text{slot-ins}} = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$

Hermes guarantees slot-level update privacy if $\mathrm{Adv}_{\mathcal{A}}^{\text{slot-ins}}$ is negligible in the security parameter λ.

Theorem 3.4. *If the underlying BFV encryption scheme is IND-CPA secure, then Hermes satisfies slot-update indistinguishability for any polynomial-time adversary $\mathcal{A}$.*

Proof Sketch. We construct a proof by contradiction showing that an adversary cannot distinguish which specific slot index was modified during an insertion or deletion. The homomorphic update logic isolates the target slot using public position-independent masks and executes a uniform sequence of Galois rotations and plaintext multiplications regardless of the target index.

Because the inserted or deleted payload is encrypted under the semantically secure BFV scheme before being homomorphically added to the masked vector, the resulting ciphertext maintains statistical uniformity. Any non-negligible advantage in guessing the updated slot would directly contradict the indistinguishability of the fresh ciphertext. The complete indistinguishability game is detailed in our technical report.

The full proof can be found in the Appendix, where we show that any adversary $\mathcal{A}$ that can distinguish between updates at different slots can be used to construct a simulator that breaks the IND-CPA security of the underlying BFV scheme, thus confirming that Hermes maintains slot-level update privacy under the given assumptions. □

It is important to note that Hermes maintains *position-hiding homomorphism* not by permuting slot values, but by maintaining the slot structure while masking all auxiliary effects of an update. The Galois shift operator is position-invariant modulo a known offset; the zeroing mask ensures no unintentional data leaks from adjacent slots. The reserved auxiliary slot (e.g., local sum) is updated using scalar plaintext operations, and its growth reflects only the net value change, not the specific position of insertion or deletion. This makes Hermes suitable for applications requiring structural privacy in addition to semantic security.

3.4.4 Correctness of Encrypted Updates and Aggregates. We now formally prove that Hermes *correctly* maintains the semantics of encrypted vector updates and aggregate computations in the presence of insertions and deletions.

Theorem 3.5. *Hermes guarantees the following three properties for any sequence of homomorphic insertions, deletions, and global aggregations over packed ciphertexts:*

(1) *The logical payload vector within each ciphertext remains consistent with the intended insertions and deletions, preserving the correct order and values of active slots.*
(2) *The auxiliary sum slot accurately reflects the current sum of all valid payload slots after any sequence of updates, ensuring that local statistics remain correct.*
(3) *The result of encrypted aggregation (e.g., 'SUM') over multiple ciphertexts yields the correct aggregate value corresponding to the underlying plaintexts, even in the presence of concurrent updates.*

Proof Sketch. The correctness of the Hermes data model is proven by algebraically tracing the homomorphic slot manipulations and verifying them against the expected plaintext relational operations. For insertions and deletions, we demonstrate that the combination of Galois shifting and zeroing masks perfectly preserves the structural order of the active payload slots while maintaining the strict capacity invariants.

Similarly, the auxiliary local sum is incrementally maintained via homomorphic scalar addition or subtraction, ensuring perfect synchronization with the payload. When global aggregation is evaluated, the homomorphic addition across the aligned auxiliary slots naturally yields the exact cumulative sum of all underlying tuples. A step-by-step formal expansion is available in the extended technical report.

The full proof can be found in the Appendix, where we provide a rigorous algebraic verification of each property, demonstrating that the homomorphic operations in Hermes faithfully implement the intended relational semantics under encryption. □

3.4.5 Computational Complexity Comparison. The primary advantage of Hermes lies in its reduction of the computational cost of relational aggregation. Table 1 provides a quantitative comparison of Hermes against scalar FHE and standard packed FHE baseline.

Table 1: Computational Complexity Comparison per Ciphertext

Operation	Scalar FHE	Standard Packed	Hermes
Global Sum	$O(n) \cdot$ Add	$O(\log n) \cdot$ Rot	$O(1) \cdot$ Add
Insertion	$O(1) \cdot$ Enc	$O(1) \cdot$ Rot	$O(1) \cdot$ Rot
Deletion	$O(1) \cdot$ Update	$O(1) \cdot$ Rot	$O(1) \cdot$ Rot
Storage	$O(n) \cdot$ CT	$O(1) \cdot$ CT	$O(1) \cdot$ CT

This asymptotic improvement is the theoretical foundation for the significant speedups observed in our experimental evaluation. By shifting the complexity from rotation-heavy query time to lightweight update time, Hermes effectively bridges the gap between relational performance and cryptographic security.

4 System Implementation

We implement Hermes as a suite of MySQL loadable functions written in C++ with OpenFHE v1.2.4. The system consists of modular components for slot-wise packing, homomorphic insertion and deletion, local-sum-aware aggregation, and slot-selective decryption, all of which are executed natively within the MySQL runtime.

As of April 2026, the codebase comprises 4761 lines of code. All components are implemented in C++ and compiled into dynamic libraries registered as MySQL loadable functions. Our minimalist integration strategy avoids any modification to the MySQL source code or query engine internals. The full source code is publicly available on Github. Experiments and development were partially supported by the NSF-funded Chameleon Cloud infrastructure [29].

More implementation details can be found in the Appendix.

4.1 Architecture and Integration

Each cryptographic operation is compiled as a C++ function into a shared object file, which is dynamically linked into the MySQL server process. This mechanism preserves full compatibility with unmodified MySQL 8.x deployments, requiring no server side modifications or query proxies. Each plugin function adheres strictly to the MySQL UDF lifecycle. The `_init` function validates argument counts and types while allocating necessary internal states. The core execution resides in `_func`, which runs per row during query evaluation. Finally, the optional `_deinit` function releases allocated memory and handles.

To bridge MySQL data types and OpenFHE plaintexts, all inputs undergo explicit parsing across the foreign function interface (FFI) boundary. MySQL UDF arguments arrive as raw C pointers tagged with type enumerations, such as `INT_RESULT` or `STRING_RESULT`. Hermes implements strict casting logic to convert these inputs into 64-bit signed integers, utilizing `std::istringstream` when numerical data is passed as string types. These scalar values are subsequently passed to the OpenFHE `MakePackedPlaintext` API.

4.2 Memory Centric Ciphertext Management

Instead of serializing ciphertexts into strings or blobs for every operation, ciphertext objects are allocated on the heap and remain within the memory space of the MySQL process. For diagnostic and linkage purposes, Hermes returns a string-encoded metadata footprint of the ciphertext rather than the serialized payload. This diagnostic string takes the format of `0xADDR (v, size=s)`, where `ADDR` is the hexadecimal memory address of the ciphertext object, $v \in \mathbb{Z}$ is a verified decrypted payload used for internal sanity checks, and s represents the object footprint in bytes.

This memory-centric layout minimizes I/O overhead during query execution. When operations like `HERMES_PACK_CONVERT()` are invoked in a query, the plugin parses the string to recover the memory address, safely casts it back to a `Ciphertext<DCRTPoly>` pointer, and performs the requested homomorphic evaluation directly in memory. This pointer-based memory lifecycle ensures that the heavy cryptographic objects never cross the FFI boundary, keeping the SQL engine lightweight and avoiding deserialization penalties.

4.3 Context and Key Management

To ensure consistent cryptographic behavior across all SQL plugin functions, Hermes constructs a single shared encryption context during the initialization phase. This context is generated using the OpenFHE `GenCryptoContext` API. We parameterize the environment with a plaintext modulus $t = 65537$, a ring dimension $N = 2^{14}$, and a multiplicative depth $d = 2$. These parameters are selected to balance computational latency with a sufficient noise budget for query evaluations.

Once the core context is established in memory, we explicitly enable the required OpenFHE operational modes, including Public Key Encryption (PKE), Leveled Somewhat Homomorphic Encryption (SHE), and Advanced SHE to unlock evaluation keys. Key generation occurs once at system startup. `KeyGen()` is invoked to produce the public and secret keys. Subsequently, Hermes generates the necessary auxiliary cryptographic materials. This involves calling `EvalMultKeyGen()` to create relinearization keys and `EvalSumKeyGen()` along with `EvalAtIndex()` to generate the Galois keys required for slot rotation algorithms. All cryptographic keys and the evaluation context are encapsulated within static C++ global variables protected by thread-safe initialization flags.

5 Evaluation

5.1 Experimental Setup

Platform. All experiments are conducted on a Dell PowerEdge R6525 server provisioned via the CHI@UC Chameleon testbed [29]. The compute node is equipped with dual AMD EPYC 7763 processors operating at 2.45GHz, providing a total of 128 physical cores and 256 hardware threads, alongside 512GB of memory. The system runs on Ubuntu 24.04 LTS and utilizes MySQL version 8.0.42 as the underlying relational database engine. The homomorphic cryptographic operations are implemented using OpenFHE version 1.2.4 configured with the BFV scheme. For all evaluations, we establish the encryption context using a default ring dimension of $N = 2^{14}$ and a plaintext modulus of $t = 2^{16}$, which yields exactly $n = 8192$ plaintext slots per ciphertext for vectorized execution.

Datasets. We evaluate Hermes on three real-world datasets that represent diverse application domains and data distributions.

- The *COVID-19* dataset [13] contains 341 daily records of pandemic-related statistics in the U.S., such as hospitalizations and case counts.
- The *hg38* dataset [25] corresponds to genomic annotations in the human reference genome, consisting of over 34,000 entries such as transcription start sites and exon counts.
- The *Bitcoin* dataset [6] records cryptocurrency trade volumes on a 3-day interval from 2013 to 2022. Since raw Bitcoin trade values are extremely large, we apply a rescaling factor of 1/24 to obtain minute-level approximations before encryption.

All datasets are stored as single-attribute tables in a MySQL database. Hermes operates over these tables using only SQL-based UDFs, without modifying schema definitions or client-side logic.

Baselines. To comprehensively evaluate the architectural and algorithmic advantages of Hermes, we establish four distinct baselines. We partition these baselines into end-to-end system macrobenchmarks executed within the database engine and isolated cryptographic microbenchmarks.

- For the end-to-end evaluations within MySQL, we compare Hermes against a Singular FHE approach and a Plaintext baseline. The Singular FHE baseline represents a naive integration where each database tuple is encrypted into an

independent ciphertext without SIMD batching. This configuration highlights the computational and storage overheads of unpaced encryption and justifies our vectorized data model. The Plaintext baseline operates on completely unencrypted data using standard SQL queries. It serves as the absolute performance ceiling to quantify the exact system-level penalty introduced by our cryptographic layer.

- To isolate our algorithmic improvements from the inherent execution overhead of the database engine, we implement two additional baselines as standalone C++ microbenchmarks. The first is a Standard Packed FHE baseline, which utilizes the same multi-slot ciphertext layout as Hermes but relies on the conventional sequence of logarithmic Galois rotations and masking operations to compute aggregate sums. This comparison isolates the latency reduction achieved specifically by our auxiliary local sum embedding technique. The second microbenchmark employs the Paillier cryptosystem [33], representing the traditional Partially Homomorphic Encryption (PHE) standard for secure database aggregations. Because Paillier supports only scalar additive homomorphism, this evaluation demonstrates how the vectorized parallelism of modern FHE outpaces legacy scalar schemes in analytical throughput.

Objectives and Metrics. We structure experiments around four research questions (RQs):

- RQ1: How does the vectorized packing data model improve initial data ingestion and encryption throughput compared to singular tuple encryption? We address this in Section 5.3.
- RQ2: What is the computational latency of performing secure in-place modifications such as record insertions and deletions over packed ciphertexts? We explore this in Section 5.4.
- RQ3: How does the auxiliary local-sum embedding technique perform against traditional cryptographic aggregation queries in isolated microbenchmarks? We answer this in Section 5.5.
- RQ4: How does the logical group capacity influence the overall execution latency across different relational operators? We analyze this in Section 5.6.

5.2 Scope of Comparison

While several homomorphic prototypes such as HE3DB [5] and ArcEDB [59] exist, we exclude them from empirical comparison for two reasons. First, these works typically function as standalone programmatic scripts rather than being integrated into the execution pipeline of a production-grade DBMS. Second, Hermes prioritizes strict IND-CPA security without any access pattern leakage, whereas those alternatives rely on interactive protocols or property-preserving encryption that inherently weaken the threat model. Thus, we benchmark against Intel IPCL [26] and OpenFHE [3] to establish a baseline using identical cryptographic security levels.

The current evaluation focuses on single-attribute analytical workloads, as these represent the primary computational bottleneck for integrating FHE into relational engines. By establishing a high-performance foundation for atomic relational operators, we demonstrate the practical feasibility of encrypted query processing.

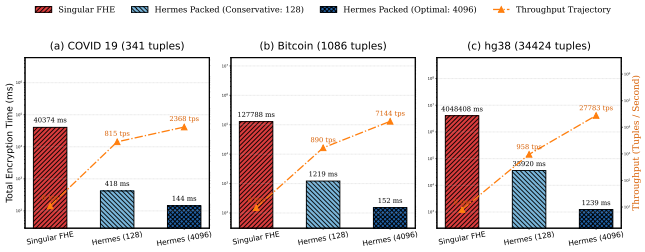

Figure 1: Encryption throughput and absolute time comparison across three datasets.

5.3 Encryption Throughput

To answer RQ1, we evaluate the initial data ingestion performance by focusing strictly on the cryptographic overhead. We compare the Hermes vectorized packing model at its boundary capacities against the Singular FHE baseline. Figure 1 presents both the absolute time required to encrypt the complete tables and the normalized throughput measured in tuples per second. The vertical axes utilize a logarithmic scale to properly depict the massive performance gap.

The Singular FHE baseline forces the database to generate an independent ciphertext for every single tuple. As shown in Figure 1(c), encrypting the hg38 dataset tuple by tuple consumes over 4048 seconds, yielding a slow throughput of merely 8 tuples per second. By contrast, Hermes leverages the SIMD capabilities of the BFV scheme to pack multiple relational tuples into a single ciphertext. Even under the most conservative structural configuration of 128 slots, the encryption time for hg38 drops to 35 seconds, boosting the throughput to 958 tuples per second.

When operating at the optimal packing scale of 4096 slots, Hermes reduces the total encryption time for hg38 to 1.2 seconds. The overlaid throughput trajectory highlights this dramatic efficiency gain, revealing a high processing rate of 27783 tuples per second, which translates to a staggering 3472x speedup over the singular encryption baseline. Similar computational boundaries are observed in the Bitcoin and COVID-19 datasets. These measurements confirm that the vectorized data model successfully reduces the fundamental throughput bottleneck of applying homomorphic encryption to relational database systems.

5.4 In-Place Update Latency

To answer RQ2, we evaluate the system performance for in-place data modifications. We focus on the amortized latency per tuple to accurately reflect the throughput capabilities of the vectorized packing architecture under bulk operation workloads.

Figure 2 presents the amortized latency for insertion operations. The Singular FHE baseline exhibits a severe bottleneck, requiring approximately 124 milliseconds to insert a single tuple. This delay stems from the mathematical requirement to generate a complete polynomial ciphertext for every isolated record. Hermes leverages the SIMD slots to distribute this cryptographic overhead. Under the optimal configuration of Hermes-4096, the amortized insertion time is reduced to approximately 0.03 milliseconds per tuple. This structural advantage translates to an over 4000x speedup for the Bitcoin dataset and similar gains across the other workloads.

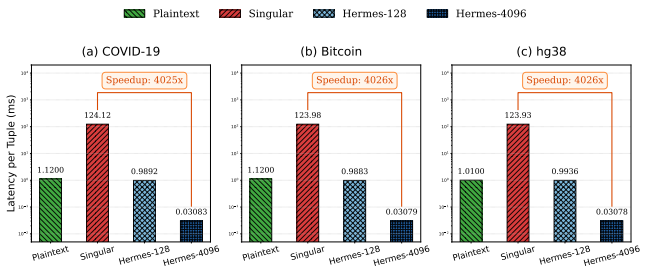

Figure 2: Amortized in-place insertion latency per tuple across three evaluation datasets.

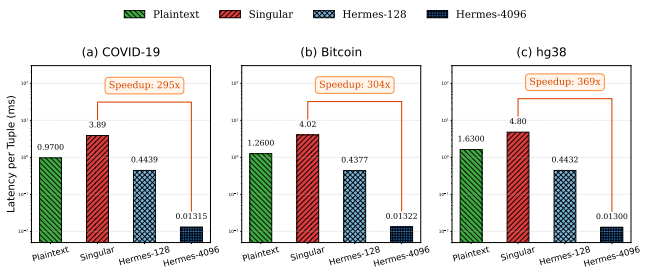

Figure 3: Amortized in-place deletion latency per tuple across three evaluation datasets.

Figure 3 illustrates the performance of deletion operations. In the Singular FHE system, a deletion is a direct SQL drop operation requiring merely 4 milliseconds. Hermes cannot physically drop rows within a packed ciphertext and must instead execute homomorphic masking to erase specific slots. Although a single packed deletion takes around 54 milliseconds to compute, amortizing this operation across 4096 slots yields an effective latency of 0.013 milliseconds per tuple. As highlighted by the overlaid metrics, this results in a 369x speedup over the singular approach for the hg38 dataset. Other two datasets show similar trends with speedups around 300x.

These empirical results validate that the SIMD packing design does not degrade the practical update performance. By amortizing the cryptographic polynomial operations across thousands of relational tuples, Hermes successfully maintains sub-millisecond effective latencies for both insertions and deletions.

5.5 Cryptographic Aggregation Queries

To answer RQ3, we evaluate the performance of Hermes on aggregation queries. To isolate the intrinsic cryptographic efficiency of our data model from the inherent execution overhead of the MySQL engine, we conduct these evaluations at the core operator level. Although Hermes natively processes SQL aggregations, measuring them end-to-end would introduce parsing and scheduling artifacts that obscure the pure cryptographic throughput. Therefore, we implement this section's benchmarks as standalone C++ programs. This approach allows us to directly compare our rotation-free aggregation technique against traditional cryptographic methods under strict computational isolation. All benchmarks are executed with $N = 2^{14}$ and $t = 65537$, yielding 8192 slots.

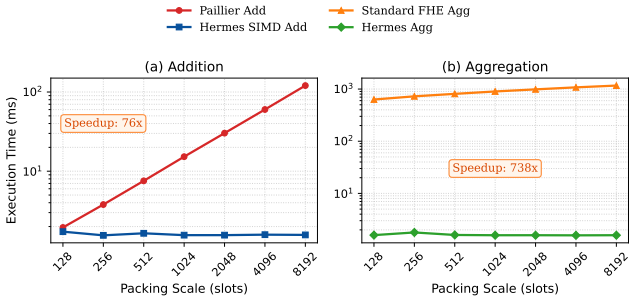

Figure 4: Cryptographic query microbenchmarks.

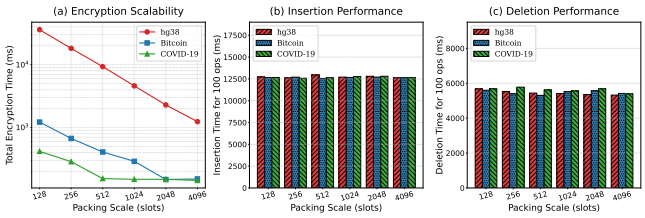

Figure 5: Performance scalability of Hermes across varying packing scales from 128 to 4096 slots.

Figure 4(a) compares the throughput of encrypted addition between the proposed Hermes data model and the Intel Paillier Cryptosystem Library (IPCL) [26]. While Paillier is a highly optimized partially homomorphic scheme, its scalar nature requires n separate ciphertext additions to process n tuples. As the packing scale increases to 8192 slots, Paillier's latency grows linearly to 120.5 ms. In contrast, Hermes utilizes the SIMD capabilities of BFV to perform a single vector-add operation that processes all 8192 slots simultaneously. This results in a near-constant latency of 1.57 ms, representing a 76x throughput advantage at the maximum scale.

Figure 4(b) evaluates the scalability of encrypted aggregation. Standard FHE aggregation strategies rely on a sequence of $\log n$ Galois rotations and relinearizations to sum internal slots. At 8192 slots, this conventional approach consumes over 1165 ms per ciphertext. Hermes fundamentally bypasses this bottleneck by embedding a precomputed local sum into a reserved auxiliary slot during the packing phase. Global aggregation in Hermes is thus reduced to a single ciphertext-level addition over these auxiliary slots. As shown in the results, Hermes completes the same aggregation in 1.58 ms, achieving a 738x speedup over the rotation-based baseline.

These microbenchmarks demonstrate that the performance gains in Hermes are not merely implementation-level optimizations but stem from a structural shift in how homomorphic primitives are applied to relational data. By aligning the database layout with the algebraic invariants of SIMD-FHE, Hermes transforms traditionally expensive cryptographic tasks into constant-time operations.

5.6 Scalability of Packing Granularity

To address RQ4, we analyze the system performance across different logical group capacities ranging from 128 to 4096 slots. Figure 5

illustrates the execution latency for encryption, insertion, and deletion operations under varying packing scales.

Figure 5(a) demonstrates a clear inverse relationship between the packing scale and the total encryption time. As the packing capacity doubles, the number of required physical ciphertexts is reduced by half. This leads to an exponential reduction in the overall cryptographic overhead. For example, encrypting the complete hg38 dataset takes over 35 seconds with a pack size of 128 but drops to approximately 1.2 seconds when the scale expands to 4096. This validates the design philosophy that maximizing the utilization of plaintext slots effectively amortizes the initial data ingestion latency.

Conversely, Figure 5(b) and Figure 5(c) reveal that the performance of in-place modifications remains entirely decoupled from the packing scale. The time required to perform 100 insertions or 100 deletions stays roughly constant across all scales. Specifically, the batch insertions stabilize at approximately 12.6 seconds, while the corresponding deletions take about 5.4 seconds. This constant time complexity is a direct manifestation of the underlying homomorphic mechanics. Because the polynomial ring dimension is fixed at $N = 2^{14}$ during the cryptographic context initialization, the computational cost of shifting and masking operations depends solely on the algebraic structure rather than the logical payload. Processing a ciphertext containing 128 active slots requires the exact same CPU cycles as processing one with 4096 active slots. Consequently, system administrators can safely maximize the packing scale to optimize storage footprints and encryption throughput without incurring any latency penalties during subsequent updates.

6 Discussion and Limitations

While Hermes demonstrates significant performance improvements for encrypted query processing, integrating fully homomorphic encryption into a relational database engine introduces several architectural trade offs. In this section, we discuss the current limitations of our prototype and outline potential avenues for future research.

6.1 Multi-Attribute Operations and Table Joins

The current implementation of Hermes focuses primarily on single attribute analytical workloads, which represent the fundamental computational bottleneck in homomorphic databases. Our vectorized packing model securely accelerates in-place updates and aggregations for isolated columns.

However, real-world queries often involve multi-attribute predicates and complex table joins. Extending Hermes to support these operations requires rethinking the packing layout. In multi-attribute tables, a columnar packing strategy could be employed where each attribute is encrypted into a separate ciphertext, maintaining a shared group ID across columns. Furthermore, executing homomorphic joins requires cross-ciphertext multiplications followed by masking, which significantly consumes the multiplicative depth (noise budget) of the FHE scheme. Future work will explore integrating specialized bootstrapping techniques or layout-aware join algorithms to support relational algebra over multiple encrypted tables.

6.2 Concurrency Control and Ciphertext Locks

A fundamental consequence of the SIMD-packed data model is that multiple relational tuples are physically collocated within a single ciphertext object. While this structure exponentially increases encryption throughput and amortizes aggregation costs, it introduces challenges for database concurrency control.

When two concurrent transactions attempt to insert or delete different tuples belonging to the same logical group, they must modify the identical underlying ciphertext. In the current Hermes prototype, we rely on the native locking protocols of the MySQL engine, which essentially escalates row-level modifications into group-level ciphertext locks to prevent race conditions. Under highly concurrent write workloads, this can lead to write amplification and lock contention. To mitigate this, future iterations of the system could implement an optimistic concurrency control (OCC) mechanism tailored for homomorphic operations, allowing non-conflicting slot updates to be temporarily buffered and merged homomorphically before a final commit.

6.3 Fragmentation and Garbage Collection

Hermes performs data-oblivious deletions by homomorphically zeroing out the targeted slots and shifting the remaining payload. As records are continuously deleted, a packed ciphertext may accumulate a significant number of stale, empty slots, leading to internal fragmentation. Although these empty slots do not affect the correctness of global aggregate queries (since their values are logically zero), they decrease the overall storage efficiency of the database.

Currently, Hermes treats these heavily deleted ciphertexts as semantically inert but retains them in storage to preserve structural consistency. To address this limitation in production environments, a background garbage collection mechanism, conceptually similar to the *autovacuum daemon* in PostgreSQL, is desired. During periods of low system load, this daemon could identify sparse ciphertexts, homomorphically compact the remaining active slots, and merge multiple underutilized groups into a single densely packed ciphertext, thereby reclaiming storage space and maintaining optimal query throughput.

7 Conclusion

This paper introduces Hermes, a prototype system designed to execute secure and efficient vectorized relational queries over homomorphically encrypted data. By incorporating a packed relational data model with embedded auxiliary aggregates, Hermes achieves constant-time encrypted aggregation while bypassing expensive cryptographic rotations. Furthermore, the introduction of data-oblivious mutability primitives allows for dynamic record-level insertions and deletions directly on packed ciphertexts via homomorphic shift and mask operations. Built seamlessly atop modern FHE schemes and deployed as native MySQL loadable functions, Hermes successfully bridges the gap between relational databases and cryptographic abstractions without modifying the core database engine. Extensive evaluations across diverse real-world datasets validate the efficacy of our approach, demonstrating massive performance gains in encryption throughput, tuple insertions, and deletions compared to state-of-the-art baselines.

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A Full Proofs

Lemma A.1. *Let $\mathcal{C}$ be the set of ciphertexts generated and manipulated by Hermes. Then under the assumption that the base FHE scheme (BFV) is IND-CPA secure, the ciphertext $c \in \mathcal{C}$ remains IND-CPA secure throughout all operations in Hermes.*

Proof. The proof proceeds by reduction. Suppose there exists an adversary $\mathcal{A}_{\text{Hermes}}$ that can distinguish two ciphertexts $c_0, c_1 \in \mathcal{C}$ produced under Hermes' operations with non-negligible advantage. Then we construct an adversary $\mathcal{A}_{\text{BFV}}$ that breaks IND-CPA security of the underlying BFV scheme.

We consider the following components:

- **Packing**: When packing multiple values into a single ciphertext, we construct a vector $v = (v_0, \ldots, v_{n-2}, s)$ where $s = \sum_{i=0}^{n-2} v_i$. The encryption $\mathsf{Enc}(v)$ is indistinguishable from $\mathsf{Enc}(v')$ for any v' of same dimension, as this reduces to IND-CPA security over vector plaintexts in BFV.
- **Auxiliary Slot**: Including a local sum s in the last slot of the plaintext vector does not affect security, since s is deterministically computed from values already encrypted. Its inclusion does not leak new information beyond what is already present in the vector.
- **Insertion**: To insert a plaintext v into a packed ciphertext c, we encrypt v into a ciphertext c_v, construct a plaintext mask m with $m[i] = 1$, and compute $c' = c + \mathsf{EvalMult}(c_v, m)$. The resulting c' remains IND-CPA secure because it is the sum of IND-CPA secure ciphertexts and homomorphic evaluations thereof. The slot mask m is public and fixed, so no leakage is introduced.
- **Deletion**: Similar reasoning applies. The rotated ciphertext and masked-out tail are both outputs of deterministic homomorphic operations on encrypted data, preserving indistinguishability under the semantic security of BFV.
- **Aggregation**: Aggregation over auxiliary slots only involves ciphertext addition. Since BFV supports additive homomorphism, and all ciphertexts involved are IND-CPA secure, the sum remains indistinguishable.

Thus, any distinguishing advantage of $\mathcal{A}_{\text{Hermes}}$ implies an advantage in distinguishing $\mathsf{Enc}(v)$ vs $\mathsf{Enc}(v')$, contradicting the IND-CPA security of BFV. □

Theorem A.2. *If the underlying BFV encryption scheme is IND-CPA secure, then Hermes satisfies slot-update indistinguishability for any polynomial-time adversary $\mathcal{A}$.*

Proof. We reduce the problem to the IND-CPA security of BFV and the semantic uniformity of the masking and shift primitives. Let m_i be a plaintext mask such that $m_i[j] = \delta$ if $j = i$, and 0 otherwise. The insertion operation computes:

$$c' = \mathsf{EvalAdd}(c_{\text{shifted}}, \mathsf{EvalMult}(c_\delta, m_i)),$$

where c_{shifted} is the result of a homomorphic rotation and masking on c that preserves all but slot i.

Because $c_\delta = \mathsf{Enc}(\delta)$ is fresh and semantically secure under BFV, and m_i is public and independent of δ, the term $\mathsf{EvalMult}(c_\delta, m_i)$ remains indistinguishable regardless of the value of i. The subsequent addition with c_{shifted} is also homomorphic and does not reveal slot index, since the masked slots are zeroed out beforehand. As a result, the output ciphertext c' is statistically indistinguishable whether insertion occurred at i_0 or i_1.

If $\mathcal{A}$ had non-negligible advantage in distinguishing i_0 from i_1, we could construct a distinguisher $\mathcal{B}$ that breaks IND-CPA security by embedding i_b into the encryption pattern. This contradicts our assumption. □

Theorem A.3. *Hermes guarantees the following three properties for any sequence of homomorphic insertions, deletions, and global aggregations over packed ciphertexts:*

(1) *The logical payload vector within each ciphertext remains consistent with the intended insertions and deletions, preserving the correct order and values of active slots.*
(2) *The auxiliary sum slot accurately reflects the current sum of all valid payload slots after any sequence of updates, ensuring that local statistics remain correct.*
(3) *The result of encrypted aggregation (e.g., 'SUM') over multiple ciphertexts yields the correct aggregate value corresponding to the underlying plaintexts, even in the presence of concurrent updates.*

Proof. We assume a ciphertext $c \leftarrow \mathsf{Encrypt}([v_0, \ldots, v_{n-2}, \sigma])$, where $v_i \in \mathbb{Z}_t$ is the plaintext payload at slot i and $\sigma = \sum_{i=0}^{n-2} v_i$ is the precomputed local sum stored in the auxiliary slot $n-1$.

(1) Correctness of Homomorphic Insertion. Let $v_{\text{new}} \in \mathbb{Z}_t$ be a value inserted into position i (with $0 \le i \le n-2$). Define the updated plaintext vector as:

$$[v'_0, \ldots, v'_{n-2}] = [v_0, \ldots, v_{i-1}, v_{\text{new}}, v_i, \ldots, v_{n-3}]$$

The homomorphic insertion process rotates the suffix

$$\{v_i, \ldots, v_{n-2}\}$$

one position to the right using a Galois automorphism, masks the insertion point i with a one-hot vector $m^{(i)} \in \mathbb{Z}_t^n$ such that $m^{(i)}[i] = 1$ and 0 elsewhere, and adds $v_{\text{new}} \cdot m^{(i)}$ via plaintext multiplication. Let $c_{\text{ins}} = \mathsf{EvalAdd}(c_{\text{shifted}}, \mathsf{EvalMult}(\mathsf{Encrypt}(v_{\text{new}}), m^{(i)}))$. Then, under correct decryption:

$$\mathsf{Decrypt}(c_{\text{ins}})[j] = \begin{cases} v_j & \text{if } j < i \\ v_{\text{new}} & \text{if } j = i \\ v_{j-1} & \text{if } i < j \le n-2 \\ \sigma + v_{\text{new}} & \text{if } j = n-1 \end{cases},$$

which matches the intended semantics. Note that the local sum is homomorphically updated by

$$\sigma' = \sigma + v_{\text{new}} \in \mathbb{Z}_t,$$

through scalar plaintext addition on the last slot.

(2) Correctness of Homomorphic Deletion. Let v_i be deleted from position i. The deletion logic performs a Galois rotation that left-shifts all slots $j > i$, then applies a mask $z \in \mathbb{Z}_t^n$ such that $z[n-1] = 0$ and $z[j] = 1$ for $j < n-1$, to zero out the stale tail. The resulting

ciphertext c_{del} decrypts to:

$$\text{Decrypt}(c_{\text{del}})[j] = \begin{cases} v_j & \text{if } j < i \\ v_{j+1} & \text{if } i \le j \le n-3 \\ 0 & \text{if } j = n-2 \\ \sigma - v_i & \text{if } j = n-1 \end{cases},$$

where σ is again updated via scalar subtraction. The vector semantics remain preserved, and auxiliary statistics remain valid.

(3) Correctness of Global Aggregation. Let $c^{(1)}, \ldots, c^{(k)}$ be a collection of packed ciphertexts, each of form:

$$c^{(\ell)} = \text{Encrypt}\left(\left[v_0^{(\ell)}, v_1^{(\ell)}, \ldots, v_{n-2}^{(\ell)}, \sigma^{(\ell)}\right]\right),$$
$$\sigma^{(\ell)} = \sum_{j=0}^{n-2} v_j^{(\ell)}.$$

Then, the encrypted global sum is computed by:

$$c_{\text{agg}} = \text{EvalAdd}(c^{(1)}, \ldots, c^{(k)}),$$

followed by slot extraction $\text{Extract}(c_{\text{agg}}, n-1)$ to isolate the final aggregated value.

The decrypted result satisfies:

$$\text{Decrypt}(c_{\text{agg}})[n-1] = \sum_{\ell=1}^{k} \sigma^{(\ell)} = \sum_{\ell=1}^{k} \sum_{j=0}^{n-2} v_j^{(\ell)},$$

which matches the true plaintext sum across all tuples. □

B Extended Implementation Details

B.1 MySQL UDF Lifecycle and Memory Handling

The integration of Hermes into MySQL relies on the User Defined Function (UDF) interface, which follows a strictly defined execution lifecycle. For each SQL function call, the MySQL engine invokes a sequence of initialization, execution, and deinitialization routines. The `_init` function is responsible for pre allocating memory and validating the input arguments provided by the SQL parser. During this phase, Hermes performs strict checks on the number of arguments and their associated types to prevent runtime segmentation faults.

The core computational logic resides in the `_func` routine, which is executed for every row in the result set. To avoid the overhead of repeated memory allocations, Hermes utilizes the `ptr` field in the `UDF_INIT` structure to maintain persistent states across row invocations. This is particularly vital for maintaining the cryptographic context and temporary evaluation buffers. Finally, the `_deinit` function ensures that all heap allocated objects, including ciphertext pointers and intermediate vectors, are properly released to prevent memory leaks in long running database sessions.

B.2 Foreign Function Interface and Data Parsing

Bridging the gap between the MySQL C based API and the modern C++17 logic of OpenFHE requires a robust Foreign Function Interface (FFI). All data entering the plugin arrives as raw C pointers within the `UDF_ARGS` structure. Hermes implements a custom parsing layer to map these inputs to internal data structures. When numeric data is passed as a string result, the system utilizes the standard `std::istringstream` object to perform safe conversion to 64 bit signed integers.

For ciphertext management, Hermes avoids the cost of serializing large objects into SQL blobs. Instead, we return a hexadecimal string representation of the memory address. This pointer based approach allows subsequent UDF calls to resolve the ciphertext object by casting the string back to a `Ciphertext<DCRTPoly>` pointer. To ensure memory safety, every pointer resolution step is accompanied by a validation check to confirm that the address resides within the expected memory segment of the MySQL process.

B.3 Cryptographic Context and Key Initialization

The security and functionality of Hermes depend on the correct initialization of the OpenFHE environment. We configure a shared encryption context during the plugin load phase to ensure that all functions operate under consistent parameters. The system utilizes the BFV scheme with a ring dimension of $N = 2^{14}$ and a plaintext modulus of $t = 65537$[cite: 1237]. These engineering choices are optimized to provide sufficient multiplicative depth for complex relational queries while maintaining sub millisecond latency for basic additions.

Key management is handled through a singleton pattern to prevent redundant key generation. At startup, the system invokes `KeyGen()` to produce the primary key pair. Subsequently, we generate specialized evaluation keys required for homomorphic operations. This includes relinearization keys by `EvalMultKeyGen()` and a set of Galois keys produced through the `EvalSumKeyGen()` and `EvalAtIndex()` methods. These keys are stored in volatile memory and protected by mutex locks to ensure thread safe access during concurrent query execution.

B.4 Deployment and Linker Configurations

Deploying Hermes on production environments like Ubuntu 24.04 LTS requires careful configuration of the dynamic linker. Since the MySQL daemon runs as a restricted service, it does not naturally inherit the user environment variables. To make the OpenFHE shared libraries visible to the `mysqld` process, we generate a systemd override file that explicitly sets the `LD_LIBRARY_PATH`. This ensures that the plugin can resolve dependencies like `libOPENFHEcore.so` at runtime.

Furthermore, all external entry points are wrapped in `extern 'C'` linkage to maintain compatibility with the MySQL symbol resolution logic. The build process is managed via CMake, which handles the static linking of core cryptographic modules while keeping the high level UDF logic relocatable. Diagnostic information and error messages are redirected to the standard error stream, which is automatically captured by the MySQL error log for offline analysis and troubleshooting.

B.5 Security Engineering and Data Obliviousness

Hermes is designed to prevent information leakage through execution patterns or side channels. The homomorphic update logic, including insertions and deletions, is implemented using a sequence

of fixed operations. For example, the shift and mask procedure for slot updates uses a constant number of Galois rotations and multiplications regardless of the target index. This data oblivious design ensures that the server cannot infer which record was modified by observing the timing or the sequence of API calls.

We also implement rigorous bounds checking for all slot based operations. Any attempt to access an index beyond the logical length of the packed ciphertext results in an immediate error return. The auxiliary metadata slot, which stores the local sum, is protected from direct user modification. These defensive measures prevent malformed SQL queries from compromising the structural integrity of the encrypted database or triggering buffer overflows.

B.6 Benchmarking and Decryption Harness

To facilitate performance profiling without the overhead of network communication, Hermes includes a benchmarking harness that allows for local decryption within the server process. This interface is strictly restricted to experimental environments and is not intended for production use. It provides dedicated UDFs like `HERMES_DEC_VECTOR_SUMMABLE()` to extract scalar results from encrypted aggregates.

The harness utilizes a resident secret key to decrypt result ciphertexts and convert them into standard C++ vectors. We optimize this process by extracting only the requested slot to minimize memory copying. The instrumentation also captures the precise execution time of each homomorphic operator, allowing us to isolate the cryptographic latency from the SQL engine overhead. This detailed profiling was essential for validating the performance gains achieved through the auxiliary sum embedding technique and the vectorized packing model.